

Models to Disentangle the Between- and Within-subjects Relationship

5.1 Introduction

In Chapter 3, regression-based methods were introduced to analyse the longitudinal relationship between a continuous outcome variable and several covariates. One of the most difficult parts of longitudinal regression analysis is the interpretation of the regression coefficient. In Section 3.3.4 it was explained that the regression coefficient derived from longitudinal regression method is a weighted average of a between-subjects and a within-subjects relationship. Although this is an important strength of the analysis, it also limits the interpretation of the result in such a way that no separation can be made between the two aspects of the longitudinal relationship. In this chapter, a few alternative models will be discussed which aim to disentangle the between-subjects and the within-subjects part of the relationship or which aim to estimate only the within-subjects part of the relationship.

5.2 Hybrid Models

Hybrid models, which are also known as between-within models, are used to disentangle the between-subjects and within-subjects part of the longitudinal relationship (Curren and Bauer, 2001). The between-subjects part of the relationship can be estimated with the relationship between the mean value of the particular covariate for each subject and the repeatedly measured outcome variable (Equation 5.1). To obtain the within-subjects part of the relationship, the covariate can be centred around the mean of the particular subject (Equation 5.2). The difference between the observations at each time-point and the individual mean value is known as the deviation score. To obtain both the between- and within-subjects part of the relationship, a combination of Equation 5.1 and Equation 5.2 can be applied (Equation 5.3):

$$Y_{it} = \beta_0 + \beta_b \bar{X}_i + \varepsilon_{it} \quad (5.1)$$

$$Y_{it} = \beta_0 + \beta_w (X_{it} - \bar{X}_i) + \varepsilon_{it} \quad (5.2)$$

$$Y_{it} = \beta_0 + \beta_b \bar{X}_i + \beta_w (X_{it} - \bar{X}_i) + \varepsilon_{it} \quad (5.3)$$

where Y_{it} are observations of the outcome for subject i at time t , β_0 is the intercept, X_{it} are observations of the covariate for subject i at time t , β_b is the regression coefficient for the between-subjects part of the relationship, $\bar{X}_i$ is the mean value of the covariate for subject i , β_w is the regression coefficient for the within-subjects part of the relationship, and ε_{it} is the error for subject i at time t .

Because the mean value over time within a subject is not correlated with the subjects deviation score (the correlation equals zero) the regression coefficient reflecting the between-subjects relationship (Equation 5.1) and the regression coefficient reflecting the within-subjects relationship (Equation 5.2) are exactly the same as the comparable regression coefficients obtained from Equation 5.3.

5.2.1 Example

To apply the hybrid model to the example dataset (i.e. the longitudinal relationship between cholesterol and the sum of skinfolds), the mean value over time for the sum of skinfolds for each subject and the deviation score at each time-point for the sum of skinfolds has to be calculated (see Table 5.1).

Output 5.1 shows the result of the linear mixed model analysis between cholesterol and the individual mean value of the sum of skinfolds in order to obtain the between-subjects part of the relationship.

From Output 5.1 it can be seen that the between-subjects part of the relationship between cholesterol and the sum of skinfolds is somewhat higher than the overall relationship. The regression coefficient for the between-subjects part equals 0.2042728, while the overall regression coefficient (see Output 3.1) was equal to 0.1871179. In Chapter 3 it has already been mentioned that the

overall regression coefficient is a weighted average of the between-subjects and within-subjects part of the relationship. This directly implies that the within-subjects part of the relationship must be less strong than the overall regression coefficient.

Table 5.1 Data structure needed for performing a hybrid model analysis

Id	Time	Covariate X	Mean_X	Deviation_X
1	1	3	5	-2
1	2	4	5	-2
1	3	5	5	0
1	4	5	5	0
1	5	6	5	1
1	6	7	5	2

Output 5.2 shows the result of the linear mixed model analysis between cholesterol and the deviation score of the sum of skinfolds in order to obtain the within-subjects part of the relationship.

From Output 5.2 it can indeed be seen that the within-subjects part of the relationship is slightly lower than the between-subjects part of the relationship, but that they are both very close to the overall relationship. This is, however, not always the case. There are examples in which the between-subjects part of the relationship has a different sign than the within-subjects part of the relationship. To illustrate that the variables reflecting the between-subjects part and the within-subjects part of the relationship are uncorrelated, Output 5.3 shows the result of the linear mixed model analysis between cholesterol and both the individual mean value and the deviation score of the sum of skinfolds.

Output 5.1 Results of a linear mixed model analysis to analyse the relationship between cholesterol and the individual mean value of the sum of skinfolds

```
Mixed-effects ML regression          Number of obs      =      882
Group variable: id                   Number of groups   =      147

                                   Obs per group:
                                   min =          6
                                   avg =         6.0
                                   max =          6

                                   Wald chi2(1)      =      29.61
                                   Prob > chi2       =      0.0000

Log likelihood = -862.49535

-----
chol |      Coef.   Std. Err.      z    P>|z|    [ 95% Conf. Interval]
-----+-----
mean_skinf |   .2042728   .037537    5.44   0.000    .1307017   .2778439
   _cons |   3.733461   .1484745  25.15   0.000    3.442456   4.024466
-----+-----

Random-effects Parameters | Estimate Std. Err.   [ 95% Conf. Interval]
-----+-----
id: Identity              |
      var(_cons) |      .2878542   .0395184   .2199452   .3767304
-----+-----
      var(Residual) |      .3012036   .015712   .2719306   .3336279
-----+-----
LR test vs. linear model: chibar2(01) = 311.24      Prob >= chibar2 = 0.0000
```

Output 5.2 Results of a linear mixed model analysis to analyse the relationship between cholesterol and the deviation score of the sum of skinfolds

```

Mixed-effects ML regression          Number of obs      =      882
Group variable: id                  Number of groups   =      147

                                   Obs per group:
                                   min =          6
                                   avg =         6.0
                                   max =          6

                                   Wald chi2(1)      =      74.31
                                   Prob > chi2       =      0.0000

Log likelihood = -840.58997

```

	chol	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
dev_skinf		.1806347	.0209543	8.62	0.000	.1395651 .2217043
_cons		4.498141	.0525641	85.57	0.000	4.395117 4.601164

```

-----
Random-effects Parameters | Estimate Std. Err. [ 95% Conf. Interval]
-----+-----
id: Identity              |
      var(_cons)          | .3605675 .047435 .2786157 .4666246
-----+-----
      var(Residual)       | .2735468 .0142693 .2469617 .3029938
-----
LR test vs. linear model: chibar2(01) = 420.05    Prob >= chibar2 = 0.0000

```

From Output 5.3 it can be seen that the regression coefficients from the analysis with both variables are exactly the same as the regression coefficients obtained from the two separate analyses.

The hybrid model can be extended with a random slope, but this can only be done for the within-subjects part of the relationship (i.e. the deviation score). The variable reflecting the between-subjects part of the relationship (i.e. the individual mean value) is not changing over time and therefore measured on subject level. Because of that, it is not possible to add a random slope for the individual mean value to the model. Output 5.4 shows the result of a hybrid model including a random slope for the deviation score.

As with all mixed model analyses, the necessity of the random slope for the deviation score can be evaluated by comparing the -2 log likelihood of the model with only a random intercept with the -2 log likelihood of the model with a random

intercept, a random slope for the deviation score and the covariance between the random intercept and random slope. This difference equals 10.7, which is statistically significant on a Chi-square distribution with two degrees of freedom. So, the final result of this (hybrid) analysis can be derived from the analysis with a random intercept and a random slope for the deviation score shown in Output 5.4.

5.2.2 Direct Estimation of the Hybrid Model

It was already mentioned that the between-subjects part of the relationship is basically nothing more than the relationship between the mean value of the particular covariate for each subject and the time dependent outcome variable. To obtain the within-subjects part of the relationship, the covariate has to be centred around the mean of the

Output 5.3 Results of a linear mixed model analysis to analyse the relationship between cholesterol and both the individual mean value and the deviation score of the sum of skinfolds

```

Mixed-effects ML regression      Number of obs      =      882
Group variable: id              Number of groups   =      147

                                Obs per group:
                                    min =          6
                                    avg  =         6.0
                                    max  =          6

                                Wald chi2(2)      =     103.93
                                Prob > chi2       =      0.0000

Log likelihood = -827.10007

-----
      chol |      Coef.   Std. Err.      z    P>|z|    [ 95% Conf. Interval]
-----+-----
mean_skinf | .2042728    .037537    5.44   0.000    .1307017   .2778439
dev_skinf  | .1806347    .0209543    8.62   0.000    .1395651   .2217043
   _cons   | 3.733461    .1484745   25.15   0.000    3.442456   4.024466
-----+-----

Random-effects Parameters | Estimate   Std. Err.   [ 95% Conf. Interval]
-----+-----
id: Identity              |
      var(_cons) | .2924636   .0395032    .2244397   .3811044
-----+-----
      var(Residual) | .2735468   .0142693    .2469617   .3029938
-----+-----
LR test vs. linear model: chibar2(01) = 346.82    Prob >= chibar2 = 0.0000

```

particular subject. The difference between the observations at each time-point and the individual mean value is known as the deviation score and reflects the within-subjects part of the longitudinal relationship. In Section 5.2.1 both variables were calculated and analysed with a linear mixed model analysis. In STATA it is, however, also possible to obtain the between- and within-subjects part of the longitudinal relationship directly with the *xtreg* procedure. Basically, the *xtreg* procedure is a linear mixed model analysis with only a random intercept. When *fe* is added to the *xtreg* procedure, the within-subjects part of the relationship is estimated, while with *be*, the *xtreg* procedure provides the between-subjects part of the relationship. It should be noted that *fe* stands for fixed effect. This is rather confusing, because in Section 3.3.3 it was mentioned that within mixed model analysis a distinction is made between the fixed part of the

regression model and the random part of the regression model. The fixed part of the regression model is the part in which the regression coefficients are given, while the random part of the model contains the random variances and the residual variance. It should be realised that in other research fields, such as econometrics, the fixed effect model refers to a model in which the within-subjects part of the longitudinal relationship is estimated (Imlach Gunasekara et al., 2014).

To illustrate the use of the *xtreg* procedures, the example used in Section 5.2.1 (i.e. the longitudinal relationship between cholesterol and the sum of skinfolds) will be reanalysed with the *xtreg* procedure. Output 5.5 shows the result of the *xtreg* procedure to analyse the between-subjects part of the relationship, while Output 5.6 shows the result of the *xtreg* procedure to analyse the within-subjects part of the relationship.

Output 5.4 Results of a linear mixed model analysis to analyse the relationship between cholesterol and both the individual mean value and the deviation score of the sum of skinfolds with a random slope for the deviation score

Mixed-effects ML regression	Number of obs	=	882
Group variable: id	Number of groups	=	147
	Obs per group:		
	min	=	6
	avg	=	6.0
	max	=	6
	Wald chi2(2)	=	77.30
Log likelihood = -821.74781	Prob > chi2	=	0.0000

chol	Coef.	Std. Err.	z P> z [95% Conf. Interval]
-----+			
mean_skinf	.2086209	.0374826	5.57 0.000 .1351564 .2820854
dev_skinf	.193318	.0278516	6.94 0.000 .1387299 .2479061
_cons	3.717184	.1482824	25.07 0.000 3.426556 4.007812

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]
-----+			
id: Unstructured			
var(dev_sk~f)	.029545	.0129097	.0125473 .0695692
var(_cons)	.2961119	.0395121	.2279683 .3846247
cov(dev_sk~f,_cons)	.0108371	.0166647	-.0218251 .0434993
-----+			
var(Residual)	.2518422	.0143999	.2251429 .2817077

LR test vs. linear model: chi2(3) = 357.53 Prob > chi2 = 0.0000			

From Output 5.5 and Output 5.6, it can be seen that the two regression coefficients are exactly the same as the regression coefficients obtained from the hybrid mixed model analysis shown in Output 5.4. The standard errors, however, are slightly different between the two procedures as well as the random intercept variance and the residual variance. It can further be seen that the *xtreg* procedure provides some additional information which is not provided by the standard linear mixed model analyses with the mean value and the deviation score. This additional information includes the explained variances of the between-subjects part and the within-subjects part of the relationship, an estimation of the residual standard deviation and the correlation

between the regression coefficient for the within-subjects part of the relationship and the corresponding random effect. However, this additional information is not very relevant for a proper interpretation of the result.

5.2.3 Hybrid Models with Categorical Time-dependent Covariates

In the example used in this chapter, the sum of skinfolds was the continuous time-dependent covariate of interest. It has been shown that in this case it is relatively easy to calculate the variables used for the hybrid model analysis (i.e. the individual mean value and the deviation score). However, when the covariate of interest is categorical,

Output 5.5 Results of a direct estimation of the between-subjects relationship to analyse the relationship between cholesterol and the sum of skinfolds

```

Between regression (regression on group means)   Number of obs   =    882
Group variable: id                               Number of groups =    147

R-sq:
  within = 0.0918                                min =          6
  between = 0.1677                               avg =         6.0
  overall = 0.1383                               max =          6

sd(u_i + avg(e_i.)) = .5854208                   F(1,145)        =   29.21
                                                    Prob > F         =   0.0000
  
```

chol	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
skinf	.2042728	.037795	5.40	0.000	.1295726	.278973
_cons	3.733461	.1494949	24.97	0.000	3.43799	4.028932

the situation is slightly more complex. Especially when the time-dependent covariate of interest consists of more than two groups. In that case, it is not possible to calculate the between- and within-subjects variables and therefore, it is necessary to use the direct estimation of the hybrid model, which was discussed in Section 5.2.2. To illustrate this, the time-dependent covariate used in the earlier examples (sum of skinfolds) is divided into three equally sized groups at each time-point (tertiles) in order to create a time-dependent categorical covariate. First a standard linear mixed model analysis was performed to analyse the overall relationship between cholesterol and tertiles of the sum of skinfolds. Output 5.7 shows the result of this analysis.

From Output 5.7 it can be seen that there are two regression coefficients for the sum of skinfolds. The first one (0.1728317) reflects the difference in cholesterol on average over time between subjects with a sum of skinfolds in the second tertile compared to the subjects with a sum of skinfolds in the first tertile. The second regression coefficient for the sum of skinfolds (0.335043) reflects the difference in cholesterol on average over time between subjects with a sum of skinfolds in the third tertile compared to the subjects with a sum of skinfolds in the first tertile. Because the categorical sum of skinfolds variable is time-dependent, both coefficients are a combination

of the between-subjects part of the relationship and the within-subjects part of the relationship. To estimate both parts of the relationship separately, two direct hybrid model analyses should be performed with the *xtreg* procedure. Output 5.8 shows the result of the linear mixed model analysis to obtain the between-subjects part of the relationship between cholesterol and tertiles of the sum of skinfolds, while Output 5.9 shows the result of the linear mixed model analysis to obtain the within-subjects part of this relationship.

From the two outputs with the between- and within-subjects parts of the relationship it can be seen that the overall relationship between cholesterol and the tertiles of the sum of skinfolds is mostly driven by the between-subjects part of the relationship and less by the within-subjects part of the relationship.

When the time-dependent covariate of interest is dichotomous, it is possible to calculate the two variables used for the hybrid model analysis and use them for estimating the between- and within-subjects part of the relationship. It should, however, be realised that the calculated variables, i.e. the individual mean over time and the deviation score, are not dichotomous anymore (see Table 5.2). Nevertheless, the result obtained from the linear mixed model analysis with the two calculated variables will be identical to the direct estimation of the two regression coefficients with the *xtreg* procedure.

Output 5.6 Results of a direct estimation of the within-subjects relationship to analyse the relationship between cholesterol and the sum of skinfolds

```

Fixed-effects (within) regression      Number of obs   =      882
Group variable: id                    Number of groups =      147

R-sq:                                Obs per group:
    within = 0.0918                      min =          6
    between = 0.1677                     avg  =         6.0
    overall = 0.1383                     max  =          6

corr(u_i, Xb) = 0.0433                  F(1, 734)       =      74.21
                                      Prob > F         =      0.0000

-----+-----
      chol |      Coef.   Std. Err.      t    P>|t|    [ 95% Conf. Interval]
-----+-----
      skinf | .1806347   .0209685    8.61   0.000    .1394692   .2218001
      _cons | 3.821948   .0804481   47.51   0.000    3.664013   3.979884
-----+-----
      sigma_u | .58419889
      sigma_e | .52337321
      rho    | .5547529   (fraction of variance due to u_i)
-----+-----
F test that all u_i=0: F(146, 734) = 7.46      Prob > F = 0.0000

```

5.2.4 Comments

It should be realised that a hybrid model only makes sense when the covariate of interest is time-dependent. When the covariate is time-independent, which is for instance the case for sex in the example used throughout the book, the regression coefficient has only a between-subjects interpretation.

It is argued that the use of hybrid models to disentangle the between- and within-subjects part of the relationship in longitudinal studies only holds when the time-dependent covariate is not increasing or decreasing over time. When the time-dependent covariate is changing over time, it is argued that the deviation score must not be calculated around the individual mean value but that it should be calculated around the individual regression line with time. Furthermore, when the data is unbalanced, i.e. when the time period and the number of repeated measurements is different for different subjects, also the between-subjects part of the relationship should be calculated in a different way, i.e. the between-subjects part of the relationship should be captured with the intercept

of an individual regression line with time when time is centred around the grand mean (Curran and Bauer, 2001). Although the calculation of the individual regression line and the deviation from that line makes sense, a comparable result may be obtained from a hybrid model adjusted for time, which is much easier to perform. See Chapter 4 for a detailed discussion of the use of the time variable in longitudinal data analysis.

5.3 Models to Estimate the Within-subjects Part of the Longitudinal Relationship

5.3.1 Introduction

As mentioned before, the standard longitudinal regression model pools together between-subjects and within-subjects relationships. In Section 5.2, hybrid models were introduced as an elegant way to disentangle the between- and within-subjects part of the relationship. Before hybrid models were introduced within medical science, other alternative models were used. These alternative

Output 5.7 Results of a linear mixed model analysis to analyse the relationship between cholesterol and tertiles of the sum of skinfolds

```

Mixed-effects ML regression          Number of obs    =      882
Group variable: id                  Number of groups =      147

                                   Obs per group:
                                   min =         6
                                   avg =        6.0
                                   max =         6

                                   Wald chi2(2)    =      20.76
                                   Prob > chi2     =      0.0000

Log likelihood = -866.13569

-----
      chol |      Coef.  Std. Err.      z    P>|z|    [ 95% Conf. Interval]
-----+-----
  skinf_ter |
      2 | .1728317   .0615525    2.81   0.005    .0521911   .2934723
      3 | .335043    .0735408    4.56   0.000    .1909058   .4791802
      |
    _cons | 4.329033   .0640156   67.62   0.000    4.203565   4.454501
-----+-----

Random-effects Parameters | Estimate Std. Err. [ 95% Conf. Interval]
-----+-----
      id: Identity         |
      var(_cons) | .3111541 .0428257    .2375857   .4075029
-----+-----
      var(Residual) | .3002016 .0157034    .2709487   .3326127
-----+-----
LR test vs. linear model: chibar2(01) = 322.27    Prob >= chibar2 = 0.0000

```

models did not intend to disentangle the between-subjects and the within-subjects part of the relationship, but rather to estimate only the within-subjects part of the relationship. In the next sections of this chapter, two of those alternative models (i.e. the model of changes and the autoregressive model) will be discussed.

5.3.2 Model of Changes

With the model of changes, it is not the actual observed values at each time-point that are modelled, but the differences between two consecutive measurements of both the outcome and the covariates (Equation 5.4):

$$(Y_{it} - Y_{it-1}) = \beta_0 + \beta_1(X_{it} - X_{it-1}) + \varepsilon_{it} \quad (5.4)$$

where Y_{it} are observations of the outcome for subject i at time t , Y_{it-1} are observations of the outcome for subject i at time $t-1$, β_0 is the intercept, X_{it} are observations of the covariate for subject i at time t , X_{it-1} are observations of the covariate for subject i at time $t-1$, β_1 is the regression coefficient for the covariate and ε_{it} is the error for subject i at time t .

Table 5.3 shows the data structure needed to perform a model of changes.

5.3.2.1 Example

Output 5.10 shows the result of the model of changes to analyse the relationship between cholesterol and the sum of skinfolds.

From Output 5.10 it can be seen that the regression coefficient for the change in sum of skinfolds

Output 5.8 Results of a direct estimation of the between-subjects relationship to analyse the relationship between cholesterol and tertiles of the sum of skinfolds

```
Between regression (regression on group means) Number of obs      =    882
Group variable: id          Number of groups   =    147
```

R-sq:	Obs per group:
within = 0.0067	min = 6
between = 0.1569	avg = 6.0
overall = 0.0845	max = 6

	F(2,144)	= 13.40
sd(u i + avq(e i.)) = .5912291	Prob>F	= 0.0000

	chol	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
skinf_ter							
2		.3093071	.1752555	1.76	0.080	-.0370986	.6557128
3		.7284806	.1408016	5.17	0.000	.4501757	1.006786
_cons		4.152687	.1003059	41.40	0.000	3.954424	4.350949

equals 0.0736776. This indicates that for each unit of change in the sum of skinfolds there is (on average over time) a change of 0.0736776 units in cholesterol. Because the coefficient deals with changes in both the outcome and the covariate it is believed that this coefficient only reflects the within-subjects part of the longitudinal relationship between cholesterol and the sum of skinfolds. This is, however, not totally true, because the estimated regression coefficient also has a between-subjects part and a within-subjects part interpretation.

An interesting finding is the magnitude of the random intercept variance, which is shown in the random part of the model of changes. This variance is about zero, which indicates that there is basically no need to adjust for the dependency of the repeated observations within the subject. The reason for this phenomenon is that in the model of changes a different outcome is used, i.e. the changes in cholesterol between subsequent measurements. In most situations, changes between subsequent measurements are not positively correlated with each other and therefore, there is basically no need for using a mixed model analysis. To illustrate this, Output 5.11 shows the observed correlation matrix for the changes between subsequent measurements in cholesterol.

The issue of these non-positive correlations in the outcome also holds for generalised estimating equations (GEE) analysis. In Section 3.4 it was mentioned that within GEE analysis an adjustment is made for the correlated observations by directly estimating the correlation between the repeated measurements. This was done by assuming a priori a certain correlation structure. One of the options to choose from was the independent structure. An independent correlation structure seemed a bit strange, because the general idea of a longitudinal regression method is to adjust for the positive correlations between the repeated observations within the subject. However, when the changes between subsequent measurements are modelled, probably the best option for the GEE analysis is the independent correlation structure. Output 5.12 shows the result of this analysis.

Because there is no missing data in the example dataset, the regression coefficient for the change in sum of skinfolds obtained from the linear GEE analysis is exactly the same as the regression coefficient obtained from the linear mixed model analysis. The difference between the two methods is the higher standard error estimated with the GEE analysis. It has previously been explained in Section 3.4 that this has to do

Output 5.9 Results of a direct estimation of the within-subjects relationship to analyse the relationship between cholesterol and tertiles of the sum of skinfolds

```

Fixed-effects (within) regression      Number of obs   =    882
Group variable: id                   Number of groups =    147

R-sq:                                Obs per group:
    within = 0.0070                      min =         6
    between = 0.1555                     avg  =        6.0
    overall = 0.0845                     max  =         6

corr(u_i, Xb) = 0.2545                F(2, 733)       =    2.58
                                      Prob > F         =    0.0768

-----+-----
      chol |      Coef.      Std. Err.      t    P>|t|    [ 95% Conf. Interval]
-----+-----
    skinf_ter |
         2 |   .1041232    .0675186    1.54  0.123   - .0284296   .2366761
         3 |   .1944942    .0857112    2.27  0.024   .0262255   .3627629
         |
        _cons |   4.398704    .0501076   87.79  0.000    4.300332   4.497075
-----+-----
    sigma_u |   .61593675
    sigma_e |   .54764707
        rho |   .55848764    (fraction of variance due to u_i)
-----+-----
F test that all u_i=0: F(146, 733) = 7.09                Prob > F = 0.0000

```

Table 5.2 Data structure needed for performing a hybrid model analysis with a dichotomous covariate

Id	Time	Covariate X	Mean_X	Deviation_X
1	1	0	0.33	-0.33
1	2	0	0.33	-0.33
1	3	0	0.33	-0.33
1	4	0	0.33	-0.33
1	5	1	0.33	0.67
1	6	1	0.33	0.67

with the robust estimation of the standard error which is used within GEE analysis.

5.3.2.2 Another Example

A very interesting example of the use of the model of changes has been given in a study also based on data from the Amsterdam Growth and Health Longitudinal Study (Twisk et al., 1998). The

purpose of that study was to investigate the longitudinal relationship between two lung function parameters: forced vital capacity (FVC) and forced expiratory volume in one second (FEV1) and smoking behaviour. One of the characteristics of these lung function parameters is that, in a relatively healthy population, these parameters are highly influenced by anthropometrics. This indicates that the between-subjects variance of these parameters is high. On the other hand, because the population is relatively healthy, the within-subjects variance is relatively low. The result of the standard longitudinal regression analysis did not show a strong relationship between lung function parameters and smoking behaviour. Thinking about the theoretical relationship between lung function and smoking, finding a weak relationship between these two parameters seems to be very strange. However, taking into account the fact that the between-subjects variance is much higher than the within-subjects variance and also that the between-subjects part of the relationship is much

Table 5.3 Data structure needed for performing a model of changes¹

Id	Time	Outcome Y	Covariate X	Change Y	Change X
1	1	3	3	1	-1
1	2	4	2	1	0
1	3	5	2	0	2
1	4	5	4	1	0
1	5	6	4	1	2
1	6	7	6	Na	Na

¹ Change is defined as the value at time t minus the value at $t - 1$.
Na = not applicable; these lines are not used in the analysis.

Output 5.10 Results of a linear mixed model analysis to analyse the relationship between the changes in cholesterol and the changes in the sum of skinfolds

```

Mixed-effects ML regression          Number of obs    =    735
Group variable: id                  Number of groups =    147

                                   Obs per group:
                                   min =         5
                                   avg =        5.0
                                   max =         5

                                   Wald chi2(1)    =    10.73
                                   Prob > chi2     =    0.0011

Log likelihood = -643.91558

-----
      delchol |          Coef.  Std. Err.   z  P>|z|  [ 95% Conf. Interval]
-----+-----
      delskinf | .0736776   .0224937   3.28 0.001   .0295907   .1177645
        _cons | .1241763   .021809   5.69 0.000   .0814315   .1669212
-----+-----

Random-effects Parameters | Estimate Std. Err.   [ 95% Conf. Interval]
-----+-----
id: Identity              |
      var(_cons) | 2.80e-25  1.08e-24   1.46e-28  5.38e-22
-----+-----
      var(Residual) | .3376566  .0176136   .3048407  .3740052
-----+-----
LR test vs. linear model: chibar2(01) = 0.00      Prob >= chibar2 = 1.0000

```

weaker than the within-subjects part of the relationship, it is not surprising that the pooled longitudinal regression coefficient is relatively small. It has already been stated that the pooled regression

coefficient is a weighted average of the between-subjects relationship and the within-subjects relationship. The weighing of the two coefficients is related to the between-subjects and the within-

Output 5.11 Observed correlation matrix of the changes between subsequent measurements in cholesterol

	delchol1	delchol2	delchol3	delchol4	delchol5
delchol1	1.0000				
delchol2	-0.3473	1.0000			
delchol3	0.0984	-0.4385	1.0000		
delchol4	-0.1393	-0.0374	-0.2804	1.0000	
delchol5	-0.0420	0.0003	-0.0037	-0.3501	1.0000

Output 5.12 Results of a linear GEE analysis with an independent correlation structure to analyse the relationship between the changes in cholesterol and the changes in the sum of skinfolds

GEE population-averaged model		Number of obs	=	735
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group:		
Family:	Gaussian		min =	5
Correlation:	independent		avg =	5.0
			max =	5
		Wald chi2(1)	=	6.72
Scale parameter:	.3376566	Prob > chi2	=	0.0095
Pearson chi2(735):	248.18	Deviance	=	248.18
Dispersion (Pearson):	.3376566	Dispersion	=	.3376566

(Std. Err. adjusted for clustering on id)

	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
delchol					
delskinf	.0736776	.0284147	2.59	0.010	.0179859 .1293693
_cons	.1241763	.0150948	8.23	0.000	.0945911 .1537616

subjects variances. Regarding lung function in this relatively healthy population, the between-subjects variance is much higher than the within-subjects variance, and the between-subjects part of the relationship is weighted much higher than the within-subjects part of the relationship. Having a very weak between-subjects relationship will lead to a weak overall relationship. So, based on this, it is not surprising that the standard longitudinal regression analysis did not show a strong longitudinal relationship between lung function parameters and smoking behaviour. The next step in the analysis was not to model the actual values of lung function at the different time-points, but rather the changes in lung function between subsequent

measurements. It should be noted that in this model the changes in smoking behaviour were not modelled but smoking behaviour was measured at $t - 1$. Table 5.4 shows the results of both analyses.

From Table 5.4 it can be seen that the model of changes revealed a strong inverse longitudinal relationship between both lung function parameters and smoking behaviour. So, although the values of the lung function parameters were not influenced by smoking behaviour, the changes in lung function parameters over time were highly influenced by smoking behaviour. This example illustrates that the modelling of longitudinal data is more complicated than the modelling of cross-sectional data.

Table 5.4 Standardised regression coefficients and 95% confidence intervals regarding the longitudinal relationship between lung function parameters (forced vital capacity (FVC) and the forced expiratory volume in one second (FEV1)) and smoking behaviour; a comparison between the standard model and the model of changes

	FVC	FEV1
Standard model	– 0.03 (– 0.11 to 0.06)	– 0.01 (– 0.09 to 0.06)
Model of changes	– 0.13 (– 0.22 to – 0.04)**	– 0.14 (– 0.25 to – 0.04)**

** $p < 0.01$.

Table 5.5 Data structure needed for performing an autoregressive model

	Id	Time	Outcome Y	Covariate X	Y at t-1	X at t-1
	1	1	3	3	Na	Na
	1	2	4	2	3	3
	1	3	5	2	4	2
	1	4	5	4	5	3
	1	5	6	4	5	4
	1	6	7	6	6	4

Na = not applicable; these lines are not used in the analysis.

5.3.3 Autoregressive Model

Another way in which to remove the between-subjects part of the relationship is to use an autoregressive model. Autoregressive models are also known as Markov models, conditional models or transition models, and an extensive amount of literature has been devoted to these types of models (Rosner et al., 1985; Rosner and Munoz, 1988; Zeger and Qaqish, 1988). The autoregression indicates that the outcome is regressed on the outcome itself, measured one time-point earlier (Equation 5.5):

$$Y_{it} = \beta_0 + \beta_1 X_{it-1} + \beta_2 Y_{it-1} + \varepsilon_{it} \quad (5.5)$$

where Y_{it} are observations of the outcome for subject i at time t , β_0 is the intercept, X_{it-1} are observations of the covariate for subject i at time $t - 1$, β_1 is the regression coefficient for the covariate, Y_{it-1} are observations of the outcome for subject i at time $t - 1$, β_2 is the autoregression coefficient, and ε_{it} is the error for subject i at time t .

In the autoregressive model shown in Equation 5.5 it can be seen that the value of the outcome variable at time t is related not only to the value of the outcome variable at $t - 1$, but also to the value of the covariate at $t - 1$. This makes sense because basically the autoregressive model deals with the changes in the outcome variable. This can be seen when Y_{it-1} is taken to the other side of the equation. The outcome then becomes $Y_{it} - \beta_2 Y_{it-1}$, which can be seen as a sort of adjusted change in the outcome variable between the two measurements. Knowing this, it makes sense that also the covariate measured at $t - 1$ is used in the analysis. The covariate at $t - 1$ is related to the adjusted change in the outcome variable from $t - 1$ to time t .

The model shown in Equation 5.5 is called a first-order autoregressive model, because the outcome variable at time-point t is only related to the value of the outcome variable at $t - 1$. In a second-order or third-order autoregressive model, the outcome variable at time-point t is also related to the value of the outcome variable at $t - 2$ or $t - 3$. Because the value of an outcome variable at each measurement is primarily influenced by the value of this variable one measurement earlier, in practice the higher order autoregressive models are not much used.

Table 5.5 shows the data structure needed to perform an autoregressive analysis.

5.3.3.1 Example

Output 5.13 shows the result of the autoregressive analysis to analyse the relationship between cholesterol and the sum of skinfolds.

From Output 5.13 it can be seen that in the fixed part of the model there are now two covariates with corresponding regression coefficients. The first one is the regression coefficient for the sum of skinfolds measured at $t - 1$. The second one is the regression coefficient for the outcome variable cholesterol measured at $t - 1$. It has already been mentioned that because an adjustment is made in the autoregressive analysis for the outcome variable measured at $t - 1$, the outcome reflects the (adjusted) change in cholesterol between two subsequent measurements. This is only slightly different from the outcome used in the model of changes, and because of that, the autoregressive model is used to obtain an estimate of the within-subjects part of the relationship between (in this example) cholesterol and the sum of skinfolds. It

Output 5.13 Results of an autoregressive linear mixed model analysis to analyse the relationship between cholesterol and the sum of skinfolds

Mixed-effects ML regression	Number of obs	=	735				
Group variable: id	Number of groups	=	147				
	Obs per group:						
	min =		5				
	avg =		5.0				
	max =		5				
	Wald chi2(2)	=	940.28				
Log likelihood = -607.7352	Prob > chi2	=	0.0000				

chol		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+							
skinft_1		.1075459	.014569	7.38	0.000	.0789893	.1361025
cholt_1		.7575473	.0301444	25.13	0.000	.6984655	.8166291
_cons		.8040143	.1248782	6.44	0.000	.5592574	1.048771

Random-effects Parameters			Estimate	Std. Err.	[95% Conf. Interval]		
-----+							
id: Identity							
	var(_cons)		3.87e-17	1.39e-16	3.38e-20	4.44e-14	
-----+							
	var(Residual)		.3059983	.0159622	.2762591	.3389389	

LR test vs. linear model: chibar2(01) = 2.3e-13				Prob >= chibar2 = 1.0000			

should be realised, however, that in the present example, it is not the changes in the sum of skinfolds that are added as the covariate of interest, but the value of the sum of skinfolds at $t - 1$. So, the result of the autoregressive model cannot be (directly) compared to the results of the model of changes, which were reported in Section 5.3.2.1.

The random part of the model shows that the random intercept variance of the autoregressive model is almost zero. This seems to be strange, because the outcome variable in the autoregressive model is cholesterol measured at the different time-points. Table 3.1 showed the within-subjects correlations for cholesterol, which are far from zero. The reason why the random intercept variance is almost zero in the autoregressive model has to do with the fact that in longitudinal regression analysis an adjustment is made for the correlated residuals, rather than for the correlated observations. Although in most

situations the structure of the correlation between the observations is comparable to the structure of the correlation between the residuals, for an autoregressive model this is not the case. In an autoregressive model part of the correlation in the observations is explained by the addition of the outcome variable at $t - 1$ to the model. In an autoregressive model, the within-subjects correlation of the residuals is therefore different from the within-subjects correlation of the observations. Because the outcome variable at $t - 1$ is strongly related to the outcome variable at time t , almost all correlations between the observed values are explained by the same variable measured one time-point earlier. As a result, the random intercept variance of the autoregressive model is reduced to almost zero (see Section 3.6.3). This also implies that when an autoregressive model is analysed with GEE analysis an independent

Output 5.14 Results of an autoregressive linear GEE analysis with an independent correlation structure to analyse the relationship between cholesterol and the sum of skinfolds

GEE population-averaged model		Number of obs	=	735
Group variable:	id	Number of groups	=	147
Link:	identity	Obs per group:		
Family:	Gaussian	min	=	5
Correlation:	independent	avg	=	5.0
		max	=	5
		Wald chi2 (2)	=	1181.72
Scale parameter:	.3059983	Prob > chi2	=	0.0000
Pearson chi2 (735):	224.91	Deviance	=	224.91
Dispersion (Pearson):	.3059983	Dispersion	=	.3059983
(Std. Err. adjusted for clustering on id)				

		Robust		
chol		Coef.	Std. Err.	z P> z [95% Conf. Interval]
-----+-----				
skinfth_1		.1075459	.0150824	7.13 0.000 .0779849 .1371069
cholt_1		.7575473	.0256546	29.53 0.000 .7072652 .8078294
_cons		.8040143	.1038861	7.74 0.000 .6004012 1.007627

correlation structure can be used. Output 5.14 shows the result of this analysis.

5.3.4 Comments

Although the magnitude of the regression coefficients for the different models cannot be interpreted in the same way, a comparison between the regression coefficients and standard errors of the different models shows directly that the results obtained from the different models are quite different. Using an alternative model (i.e. a model of changes or an autoregressive model) can lead to different conclusions than when using the standard model. On the one hand this is strange, because all analyses attempt to answer the question of whether there is a longitudinal relationship between a continuous outcome and a particular covariate. On the other hand, however, with the different models, different parts of the longitudinal relationships are analysed, and the results of the models should be interpreted in different ways. To obtain the best answer to the question of whether there is a longitudinal relationship between the outcome variable and a covariate, maybe the results of several models should

be combined (Twisk, 1997). In practice, however, this almost never happens: a priori the most appropriate model is chosen (usually the standard model), and only that result is reported.

It is sometimes suggested to use the model fit statistics to decide which model should be used. However, when deciding which model should be used to obtain the best answer to a particular research question, comparing the fit of the models will not provide much interesting information. First of all, in the model of changes and the autoregressive model, less observations are used than in the standard model. The problem is that the number of observations highly influences the likelihood of a particular analysis and therefore all model fit statistics. Secondly, looking at the fit of the models it is obvious that, for instance, an autoregressive model provides a much better fit than a standard model. This is due to the fact that a high percentage of variance of the outcome variable at time t is explained by the value of the outcome variable at $t - 1$. This can, for instance, be seen from the values of the scale parameter presented in the GEE output and the log likelihood presented in the output of the mixed model analysis. Both values are much lower in the

autoregressive model than in the standard model. However, this does not mean that the autoregressive model should be used to obtain the best answer to the question of whether there is a longitudinal relationship between the outcome variable and the covariate. In general, it should be realised that it is better to base the choice of a specific longitudinal model on logical considerations instead of statistical ones.

It has already been mentioned that with the model of changes and with the autoregressive model an attempt is made to estimate only the within-subjects part of a relationship. It is therefore surprising that the results of the longitudinal analyses with the model of changes and the autoregressive model are quite different. One reason for the difference in results is that both alternative models use a different model of change. This can be explained by assuming

a longitudinal study with just two measurements. In the autoregressive model, $Y_2 = \beta_0 + \beta_1 Y_1$, while in the model of changes, $Y_2 - Y_1 = \beta_0$ (where β_0 is the difference between subsequent measurements), which is equal to $Y_2 = \beta_0 + Y_1$. The difference between the two equations is the coefficient β_1 . In the model of changes the change is a fixed parameter, while in the autoregressive model the change is a function of the value of Y_1 . Another reason for the difference in result between the model of changes and the autoregressive model is the different modelling of the covariate. It has already been mentioned that for the model of changes, the changes in the time-dependent covariate were also modelled. In the autoregressive model, however, the covariate measured at $t - 1$ was used. It is obvious that different modelling of the covariates (can) lead to a different result.